Guide 1. APT Project Definition Capstone Course

1. Personal Information

The following table must be completed with the requested information.

Student Name	Josefa Vallejos Sánchez
National ID (RUT)	19.290.851-5
Degree Program	Informatics Engineering (Ingeniería en Informática)
Campus	Padre Alonso de Ovalle

2. APT Project Description

In the description, briefly state the name of your APT project and the graduate profile competencies you will put into practice. If your degree program defines performance areas, also mention which performance areas the project is linked to.

Project Name	*Personal Data Compliance and Protection: Law 21.719 Framework*
Performance Area(s)	*Requirements Design and Management, IT Project Management, Computer Systems Security*
Competencies	• *Design and adapt requirements management processes, using cutting-edge methodologies and industry standards, to develop complex, innovative and high-quality IT solutions.* • *Evaluate and manage projects throughout their life cycle, according to good practices and using software methodologies and tools, to meet the organization's requirements in traditional and agile contexts.* • *Design software solutions covering their entire life cycle, according to standards, frameworks and regulations, technologies and methodologies that promote innovation, focused on the quality, security and sustainability of the project.*

3. APT Project Rationale

The following fields must be completed with the requested information. This section asks you to describe your project in detail and to justify its relevance and pertinence.

Relevance of the APT Project	*This project arises from a real and current problem in Chile: organizations lack clear processes to manage people's ARCOP rights (Access, Rectification, Cancellation, Opposition and Portability) over their personal data. It is common for a person to request the deletion of their information and for that request never to be carried out, leaving sensitive data exposed indefinitely with no transparent tracking channel.* *This topic is especially relevant to the Informatics Engineering job market because, with Law No. 21.719 on Personal Data Protection coming into force, all organizations operating in Chile (companies, foundations, educational institutions, SMEs) will have to adapt their internal processes to comply with it, creating real demand for professionals able to translate legal requirements into concrete technical and methodological solutions.* *The situation is set in the Chilean national context and potentially affects anyone who has given their data to an organization, regardless of their socioeconomic status or level of vulnerability. The value of the project is to give organizations a clear, applicable methodological tool that lets them comply with the law in an orderly way, reducing the risk of penalties and, above all, effectively protecting people's privacy.*
Description of the APT Project	*The aim is to design a Personal Data Compliance and Protection Framework that allows an organization to implement, in an orderly and verifiable way, what Law No. 21.719 requires.* *The project will be addressed by surveying the law's regulatory requirements, modeling the ARCOP request management processes in BPMN notation, preparing a risk matrix, and building a cybersecurity audit checklist, validating the whole set by applying it to a simulated case study.*
Relevance to the Graduate Profile	*The project is directly related to the Graduate Profile of Informatics Engineering, particularly the competencies of designing and adapting requirements management processes, evaluating and managing IT projects, and designing solutions focused on quality, security and sustainability.* *These competencies are essential to solve the problem described: without requirements management the law cannot be translated into operational processes; without project management its implementation cannot be structured; and without a security and regulatory compliance approach, the framework would not fulfil its central purpose of protecting personal data.*
Relation to Professional Interests	*This project responds directly to my professional interests: requirements management, agile leadership as a Scrum Master, and information security. Developing it allows me to work in these three areas at the same time and to start building applied experience in the direction I want to pursue professionally, in a leadership role in cybersecurity and personal data protection.* *Carrying out this project contributes directly to my professional development by creating a concrete, verifiable record of my ability to address a real regulatory problem with a technical and methodological solution appropriate to my degree.*

Feasibility of the APT Project	*It is possible to develop this project within the course considering the following:* 1. ***Semester Duration:*** *The project runs from August to November, with December reserved for the defense, which gives enough time for a methodological scope rather than complex software development.* 2. ***Hours Assigned to the Course:*** *By organizing the work into fortnightly sprints with concrete, bounded deliverables, the work is spread manageably within the hours of the Capstone course.* 3. ***Required Materials:*** *No expensive software licenses or complex technical infrastructure are needed; free tools for BPMN modeling (e.g., draw.io, Bizagi), spreadsheets for the risk matrix and the checklist, and the public text of Law No. 21.719 and its regulation are enough.* 4. ***External Factors That Facilitate Its Development:*** *The law is already published and in force, so the project's regulatory basis is stable and publicly available, without depending on uncertain or constantly changing sources.* 5. ***External Factors That Could Hinder Its Development and How to Address Them:*** *Since this is an individual project, there is a risk of work overload at certain stages; this is mitigated by organizing the work into sprints with a review buffer built in at the end of the work plan, which absorbs delays without compromising the November delivery date.*

A. PART II

4. Objectives

In this section you must define the general and specific objectives of the APT Project. The objectives should be stated clearly and concisely, without further explanation, so that they are understood on their own. It is suggested to write them using an infinitive verb, since this forces you to specify concrete actions.

General Objective	*Design a personal data implementation and governance framework that allows organizations that process customer or user data to meet the requirements of Law No. 21.719 on Personal Data Protection, through verifiable protocols, processes and cybersecurity controls.*
Specific Objectives	*1. Survey and systematize the regulatory requirements of Law No. 21.719, translating them into clear operational processes that an organization can follow to comply with ARCOP rights (Access, Rectification, Cancellation, Opposition and Portability).* *2. Design process flows in BPMN notation that model the handling of data subjects' requests, from receipt to resolution within the legal deadlines.* *3. Prepare a risk matrix that identifies and prioritizes the most common vulnerabilities in the processing of personal data within an organization, linking each risk to a mitigation control.*

	4. Define a cybersecurity audit checklist that allows an organization to verify its level of compliance with the law, identifying gaps and corrective actions. *5. Validate the framework by applying it to a case study or simulated scenario, assessing its clarity, applicability and practical usefulness for a real organization.*

5. Methodology

In the following section you must describe the methodology, specific to your discipline, that you will use to solve the APT project described above, including the stages and working methods.

Description of the Methodology
A mixed working methodology is proposed, combining an agile approach (Scrum adapted to individual work) for project management and progress. It allows incremental progress, with concrete and verifiable deliverables at each stage, reducing the risk of work piling up toward the end of the semester. ***1. Project Management:*** *Work is organized in two-week sprints, each tied to a specific objective, with verifiable partial deliverables at the close of each one. A Kanban board (Trello or Jira) is used to visualize progress and monitor compliance with deadlines against the course schedule.* ***2. Regulatory Requirements Gathering (Objective 1):*** *Documentary analysis of the text of Law No. 21.719 and its regulation, complemented with international reference standards (ISO/IEC 27001) and guidelines from Chile's Personal Data Protection Agency.* ***3. Process Modeling (Objective 2):*** *BPMN 2.0 notation to diagram the ARCOP request management flows, validating each process against the deadlines and conditions required by law.* ***4. Risk Management (Objective 3):*** *Risk analysis based on recognized standards (ISO 31000 or NIST), identifying threats, assessing likelihood and impact, and linking each risk to a concrete mitigation control.* ***5. Audit and Control (Objective 4):*** *Construction of the checklist from the controls defined in earlier stages, in a verification format (Compliant / Non-compliant / Observations).* ***6. Validation (Objective 5):*** *Application of the complete framework to a simulated case study, obtaining feedback (own or from the reference instructor) to adjust and validate its practical usefulness before the project closes.*

6. Evidence

Below, describe which evidence will be assessed in the progress report and in the final report of your APT project. This evidence must be agreed with your instructor. Evidence is understood as the products developed during the project whose purpose is to make visible or document how the work has been carried out.

Type of evidence (progress or final)	Name of the evidence	Description	Justification
Progress	*Kanban board screenshots*	*Screenshots of the board (Trello/Jira) showing the status of tasks at the close of each sprint.*	*Allows verifying compliance with the work plan deadlines.*
Progress	*Fortnightly progress log*	*Record of what was done, what is pending and the adjustments to the work plan.*	*Shows project follow-up and continuous improvement.*
Progress	*Records of meetings with the reference instructor*	*Date, topics discussed, agreements and observations received.*	*Supports the scope decisions and the feedback received.*
Progress	*Systematization document of Law No. 21.719*	*Table article → requirement → responsible party, and record of the sources consulted (law, regulation, Agency guidelines).*	*Evidence for Objective 1: regulatory requirements surveyed and traceable.*
Progress	*BPMN 2.0 diagrams of the ARCOP processes*	*Diagrams exported as an image and as the editable file (draw.io, Bizagi or Lucidchart); preliminary and final version of at least one.*	*Evidence for Objective 2; shows iteration and improvement of the modeling.*
Progress	*Risk matrix*	*Complete matrix (likelihood, impact and mitigation control) with a brief justification of the methodology (ISO 31000 or NIST).*	*Evidence for Objective 3: risks prioritized and linked to controls.*
Final	*Cybersecurity audit checklist*	*Final version in an applicable format (compliant / non-compliant / observations) and a record of at least one review iteration.*	*Evidence for Objective 4: verification tool usable by an organization.*
Final	*Case study and validation*	*Description of the simulated case, result of applying the complete framework (requirements, BPMN, matrix and checklist) and reflection on the findings.*	*Evidence for Objective 5: validation of the framework's practical usefulness.*
Final	*Final report and defense presentation*	*Consolidated APT project report and the presentation used in the oral defense.*	*Documents the closure and results of the project.*

7. Work Plan

In the following table, define the planning of your APT Project as required.

Competency	Activities	Description of Activities/Tasks	Resources	Duration	Responsible	Notes
IT Project Management	Sprint 0 - Foundations	Formally close Phase 1: confirm the reference instructor's reply, record in the report the description, competencies, interests, feasibility and objectives, and create the Kanban board.	Trello or Jira	Sep 2 – 8	J. Vallejos	Difficulty: no reply yet from the reference instructor. Facilitator: scope already defined in the Phase 1 reflection.
Requirements Design and Management	Sprint 1 - Regulatory requirements (Obj. 1)	Read and systematize Law No. 21.719 (ARCOP rights, deadlines, penalties and obligations), review the Agency's guidelines and translate each legal requirement into an operational requirement (article → requirement → responsible party).	The law and its regulation	Sep 9 – 22	J. Vallejos	Facilitator: the law is published and publicly available. Difficulty: length and legal language.
BPM	Sprint 2 - BPMN modeling (Obj. 2)	Diagram the flow for receiving and handling an ARCOP request (one diagram per right or a general one with variants) and validate that the timings respect the legal deadlines.	draw.io or Bizagi	Sep 23 – Oct 6	J. Vallejos	Difficulty: deciding between several diagrams or one general diagram with variants. Facilitator: command of BPMN.
Risk Management	Sprint 3 - Risk matrix (Obj. 3)	Identify typical personal data processing risks (data leaks, improper retention, lack of consent, missed deadlines), assess likelihood and impact, and link each risk to a mitigation control.	ISO 31000 or NIST (as reference).	Oct 7 – 20	J. Vallejos	Facilitator: command of risk management stated in the self-assessment.
Defensive Cybersecurity	Sprint 4 - Audit checklist (Obj. 4)	Build the checklist from the requirements (Sprint 1) and the controls (Sprint 3), in a compliant / non-compliant / observations format, usable by a person without deep technical knowledge.	Verification Spreadsheet ISO/IEC 27001	Oct 21 – Nov 3	J. Vallejos	Depends on the deliverables of Sprints 1 and 3.
Software Quality	Sprint 5 - Case study and validation (Obj. 5)	Build a simulated case study (a fictitious organization or a well-known public case in Chile), apply the complete framework and document its clarity and applicability.	Complete Framework	Nov 4 – 10	J. Vallejos	Dependence on real personal data is avoided.
IT Project Management	Sprint 6 - Consolidation	Combine the deliverables into the APT report, write the conclusions and projections of the framework, and review the format against the course guidelines.	Word Processor Official Guidelines	Nov 11 – 17	J. Vallejos	Deliverable: first complete version of the report.
IT Project Management	Buffer - Review and adjustments	Send the report to the reference instructor for feedback and incorporate the corrections. Absorbs delays from earlier sprints without compromising the November delivery.	Report	Nov 18 – 23	J. Vallejos	Safety margin against delays.
Effective Communication Skills	December - Defense	Build the project presentation, rehearse the oral defense anticipating questions about each objective, and review the full report.	Presentation Report	December	J. Vallejos	Deliverable: presentation and defense rehearsal.

8. Gantt Chart

Find a Gantt chart format that suits you and organize in it the activities planned in the previous point, considering the period assigned for developing your APT Project. You must keep the timing of the academic period across the three phases of the Portfolio Course.

Activity	Phase 1				Phase 2											Phase 3		
	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15	W 16	W 17	W 18
Sprint 0 – Foundations																		
Sprint 1 – Regulatory requirements (Obj. 1)																		
Sprint 2 – BPMN modeling (Obj. 2)																		
Sprint 3 – Risk matrix (Obj. 3)																		
Sprint 4 – Audit checklist (Obj. 4)																		
Sprint 5 – Case study and validation (Obj. 5)																		
Sprint 6 – Consolidation																		
Buffer – Review and adjustments																		
December – Defense																		